

Grade the Ripple, Not the Stone

Why AI benchmarks die, what survived a red-teamed search for new ones, and a live pilot of DEBRIEF, a benchmark that scores what a model's note changes in another model

Benchmark research project (multi-agent study coordinated with Claude Code)

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Abstract

We asked what makes an AI benchmark durable and discriminative, then searched for a new one. Seven evidence panels (about 465 fact-checked claims) covered benchmarks humans still win, benchmarks AI wins that still separate models, dead and saturated benchmarks, games, capability gaps, test-time learning and evaluation methodology. The common hypothesis that failed benchmarks “lack novel thinking a model could not have trained on” is only partly right: novelty prevents item contamination, but novel benchmarks such as ARC-AGI-3 (under 1% to 99.9% in about five months) and FrontierMath Tier 4 (5% to 98% in under 14 months) still fell. What failed benchmarks share is a cheap path to a higher score that bypasses the named ability, plus no owner renewing the task faster than labs optimise against it. We generated 47 candidates, attacked them with seven independent red teams, and ran blinded pilots on five finalists with four Claude model tiers. Pilots killed three finalists outright: clean visual rule induction, patch auditing and self-knowledge quizzes all saturated at the mid tier. A bot-writing league on a secret new game (Season Forge) produced the widest separation (0.41 to 5.08 ladder rungs). We then built and piloted DEBRIEF, which scores the share of a fixed junior model's errors that a 150-word note from the model under test removes on unseen cases. Version 0 failed its first gate: the Haiku junior priced generated tariffs at 98.6% accuracy, leaving nothing to coach. A harder v0.2 produced a coachable junior: all four coaches independently diagnosed a misconception the authors' mutation library had missed (explaining 18 of 22 practice errors), and in their first post-test replicate every coach note beat the placebo and template baselines (net fix rates of 45 to 91% against 0 to 18%). But the same notes swung by more than 100 points between replicates, so the instrument cannot yet rank models. The broad lesson is that durable benchmarks need renewal as the mechanism, an outcome measured on something other than the model's own answer, and pre-registered gates that are allowed to fail.

1 Introduction

Benchmarks are how the field decides which models are good. Most of them stop working within two years. Static sets now stay in lab headline tables for roughly 7 to 30 months, and the hardest 2025 sets went from single digits to near ceiling in 13 to 18 months. This paper has three parts:

1. **Why benchmarks succeed or fail**, from fact-checked evidence (Section 2).
2. **A red-teamed search for a new benchmark or game**, with blinded pilots that killed most of the shortlist (Section 3).
3. **DEBRIEF**, a non-game benchmark that grades the effect of a model's explanation on another model, and its live pilots, including a failed first version (Sections 4 and 5).

We use two archetypes throughout. In **A1**, humans clearly outperform AI (the ARC-AGI pattern). In **A2**, AI clearly outperforms typical humans but the results still separate strong from weak models, ideally in informative ways

such as cheap models beating expensive ones [the StudentBench pattern; Northcutt et al. (2026)].

All evidence files, fact-check logs, pilot materials and decrypted answer keys are in the repository under `research/research_notes/Novel AI benchmark and game ideas/ and results/`. Claims we could see only through search-engine summaries, because the environment blocked several primary hosts, are marked [S].

2 Why benchmarks succeed or fail

2.1 The novelty hypothesis

The working hypothesis was that failed benchmarks “lack novel critical thinking that a model can’t have been trained on beforehand”. Panel C coded 36 benchmark histories. Under the reading “unseen instances”, 5 cases support it, 10 contradict it and 21 are neutral. Under the reading “novel skills”, it is about 9 / 6 / 21. The counts depend on coding, so the verdict rests on counterexamples that hold under every reading:

- **ARC-AGI-3**, a novel interactive family with no item leakage, went from under 1% at launch (25 Mar 2026) to 62.7% on ARC’s standard harness and 98.6% on a state-preserving harness for the same model at the same effort, and 99.9% at higher effort, in about five months (ARC Prize Foundation 2026).
- **FrontierMath Tier 4**, built from unpublished problems, went from 5% to 98% in under 14 months, on an error-corrected v2 set (Epoch AI 2026).
- **ARC-AGI-1** had 49% of its private set solved by a 2020 ensemble of brute-force program searches (Chollet et al. 2024).

The converse also fails: GPQA (Rein et al. 2024), SWE-bench and AutomationBench (Zapier 2026) stayed useful without novel skills. The kernel of the hypothesis is real for static public items. AIME 2024 scores run above what AIME 2025 predicts (Balunović et al. 2025), and OpenAI retired SWE-bench Verified after models reproduced gold patches (OpenAI 2026).

Refined statement. A benchmark fails when the cheapest way to raise its score stops running through the named capability, and no owner renews items, families and protocol faster than labs optimise against them.

2.2 Failure modes, and which ones novelty prevents

Failure mode	Does novelty prevent it?	Example
Item contamination	Yes	AIME 2024; SWE-bench Verified
Training on the task family	Only while the generator stays secret	Logic-RL reached 0.99 after about 5k synthetic puzzles (Xie et al. 2025)
Brute force or writing a solver	No	ARC-AGI-1's 49% ensemble
Harness capture	No	ARC-AGI-3: 62.7% vs 98.6%, same model and effort
Broken validity	No; novel expert sets are more error-prone	A do-nothing agent scores 38% on τ -bench (Zhu et al. 2025); 42% of FrontierMath problems needed fixes
Gaming and selective submission	No	27 private Llama 4 variants on LMArena (Singh et al. 2025)
Noise and thin baselines	No	Median human baseline is 8 people (Wei et al. 2025); a 3-point gap needs about 1,000 items (Miller 2024)
Adoption failure	No	OfficeBench (Wang et al. 2024): big gap at launch, no maintainer

2.3 What durable benchmarks have in common

Twenty design principles came out of the synthesis (phase2/design_principles.md). The five with the most support:

1. **Renew task families, not just items**, on a cadence, under a named owner. ARC's benchmark lifetimes fell from about five years to about one year to about five months.
2. **Treat any public generator as future training data**. Keep scored seeds and rule sets secret and rotating.
3. **Freeze and version one harness**, and publish the gap to a bring-your-own-harness track.
4. **Seal execution**: no network during the run, grader outside the agent's reach, and release gates where a reference solution passes and a do-nothing agent fails.
5. **Baseline humans properly**: a defined population, matched interface and effort, and a sample sized by power analysis.

Where the field still has signal (Panels A, B, E):

- **Human-over-AI gaps** are now scarce and concentrated in perception, intuitive physics, exploration, real-time play and learning across episodes. Most text-only reasoning gaps have closed: ARC-AGI-1 and -2, SimpleBench and OSWorld are all at or above their human baselines.
- **Model-versus-model spread** is widest on long horizons, calibrated abstention, conduct, and outcomes that are uncapped (money earned) or measured on someone else (a student's learning). StudentBench is instructive: its expert reviews separated tutors and showed a cheap Sonnet configuration beating a pricier Gemini one, but measured learning did not separate them (0 of 364 cells significant) (Northcutt et al. 2026).

3 A red-teamed search, and what the pilots killed

3.1 Process

- Four ideation panels produced 52 ideas, merged into 47 neutral candidate specifications.
- Five independent red teams (trainability, separation, baselines and scoring, prior art and interest, holistic) scored every candidate without seeing the ideators’ reasoning: 235 verdicts in total, 53 keep, 142 revise and 40 kill.
- A revision panel cut 16 candidates, merged 6 and redesigned 7 finalists.
- Two fresh reviewers then attacked the revised designs: one tried to game them, one judged measurement value.
- We built blinded pilots of five finalists. Each answer key was encrypted with a passphrase never written to the repository. Claude Haiku, Sonnet, Opus and Fable attempted each pilot as agents, and keys were decrypted only after solving.

3.2 Pilot results

Finalist	Archetype	Pilot	Result
Hidden-Rule Lab	A1 claim	8 visual rule-induction problems, image only, no code	Sonnet and Opus 64/64, with every hidden rule stated exactly; Haiku 31/64 (chance)
Patch Auditor	A2	20 patches, 6 spec violations, tools on	Fable, Opus and Sonnet 1.000; Haiku 0.444; a fuzzing script also 1.000
Self-Knowledge Exam	A2	60 items with self-forecasts	Sonnet, Opus and Fable 60/60, so calibration only rewarded confidence near 1; deliberate failure raised scores
Season Forge	A2, complex game	Write a bot for a new fog-of-war game; rated against a sealed 4-bot ladder	Haiku 0.41 rungs; Sonnet 0.81 and Fable 0.79 (interrupted snapshots); Opus 5.08, beating the strongest sealed bot 87% of the time
Unrun Lab	A2	Forecast interventions in 5 secret simulators	Only Haiku finished: skill 9.6, below a naive extrapolator’s 16.0

Three lessons follow:

- **Clean, static reasoning tasks saturate at the mid tier.** The provisional favourite, Hidden-Rule Lab, assumed a human advantage that no longer exists for rendered scenes.
- **“Tools on” turns many audits into script problems.** Whenever the benign class is “behaviour unchanged”, differential fuzzing separates the classes without understanding.
- **Steep, novel, adversarial environments separate models widely.** Season Forge did so at modest scoring cost: 1,600 games ran in 163 seconds on four cores.

The final ranked shortlist from this search was:

1. Season Forge (A2, game).
2. Stump Arena (A1): crowd-authored items people solve and models fail, with a “cost to stump model X” headline.
3. Unrun Lab (A2).
4. Compaction Chronicle (A2): memory half-life over two-million-token streams, conditional.

Details are in `research/reports/Novel AI benchmark and game ideas.md`.

4 DEBRIEF: grade the ripple, not the stone

A parallel design track looked for a non-game benchmark. It produced five proposals, all of which were then checked against the evidence above:

- **PREMORTEM**: predict your own failures.
- **STRAIT**: straight-through rule application.
- **WEDGE**: produce an input where two rulebooks differ.
- **SECOND READER**: find planted errors.
- **DEBRIEF**: coach a junior model with one note.

The pilots above weigh against PREMORTEM (the Self-Knowledge Exam failure), WEDGE and SECOND READER (Patch Auditor’s saturation under tools), and STRAIT (static rule application saturates). DEBRIEF was chosen because it scores an outcome that is not the model’s own answer.

4.1 Task

1. A fixed, cheaper **junior** model attempts practice cases under a rulebook it has never seen, and gets some wrong.
2. The **model under test** reads the rulebook, the junior’s worked attempts and the answer key, and writes **one note of at most 150 words**.
3. The junior, given the note, answers **fresh cases the model never saw**.

The headline is the **Net Fix Rate (NFR)**: the share of the junior’s control errors on fresh cases that the note removes, net of correct answers it breaks:

$$\text{NFR} = 100 \times \frac{\sum_{ij} (p_{ij} - c_{ij})}{\sum_{ij} (1 - c_{ij})}$$

Here c_{ij} is the junior’s control correctness on case j of item i , averaged over replicates, and p_{ij} is its correctness with the note. Two baselines price the cheap paths:

- a fixed **placebo** note (“re-read every rule”);
- a programmatic **template coach** that restates, verbatim, the provisions a mutation search blames for the junior’s errors.

Confidence intervals come from a cluster bootstrap over items.

Why it resists the failure modes above:

- **Items are generated by code from a private grammar.** Every answer comes from an exact interpreter, and no LLM writes test content.
- **Code cannot compute the score**, because scored cases are hidden from the model under test and the key to the practice cases is supplied.
- **Generic advice is priced** by the placebo arm.
- **Over-general fixes are penalised**, because broken correct answers are subtracted.

The closest prior work grades teaching by dialogue or by a teacher’s explanation of single instances ([Saha et al. 2023](#); [Shi et al. 2025](#); [Teach2Eval authors 2025](#)). DEBRIEF differs in using one note, a programmatic world, fresh-case transfer and net-of-harm scoring.

4.2 Generator

Version 0.1 generates courier tariffs at three levels. The traps are:

- weights rounded up;
- volumetric weight for parcels only;
- a weekend surcharge on the base price only;
- remote-area fees by postcode prefix;
- a GOLD discount with a zone exclusion;
- a document price cap.

Each trap doubles as a misconception in a mutation library. A mutation search labels each junior error as *diagnosable* (some set of misconceptions reproduces it) or a *slip*. Distractor sections pad the specification. An offline simulation with programmatic juniors checked the instrument: empty notes score about 0%, placebo about 3%, an over-general fix about 25%, and a strong coach 75 to 80%. Confidence intervals shrink from ± 13 to ± 5 points as items go from 24 to 200.

5 DEBRIEF live pilots

We ran two live pilots with Claude models as Claude Code agents.

- **Junior:** Claude Haiku, told not to use tools, answering every case in a single reply.
- **Coaches (models under test):** Haiku, Sonnet, Opus and Fable, each writing one note per item without tools.
- **Blinding:** answer keys were held outside the repository until scoring.

Everything is in `results/pilot_v0/` and `results/pilot_v1/`: prompts, replies, notes, keys and scores.

5.1 Pilot v0 (levels L2–L3): no headroom

Six items, 8 practice and 12 fresh cases each. The junior priced **48 of 48 practice cases** correctly. On the fresh cases it scored **98.6%** in both control replicates and 100% with the placebo note. With two errors in 144 control

answers there was nothing to coach: NFR is undefined in practice, and the placebo’s nominal “100%” was two random slips disappearing.

Gate G1 failed. Pre-registered, it required control accuracy between 30% and 70%. This is the same saturation pattern seen in Section 3: a current small model already applies clearly written rule systems almost perfectly.

5.2 Pilot v0.2 (levels L4–L5): coaches diagnose well, the junior is unstable

What changed. We added interacting, counter-default provisions:

- heavy documents reclassified as parcels;
- a per-kg uplift above a weight threshold;
- discounts that exclude flat fees;
- oversize fees that double when two or more sides are long;
- a minimum charge;
- buried “amendments” that override earlier sections.

Items are larger: 10 to 12 practice cases and 14 to 16 fresh cases. We ran six items, three at L4 and three at L5. Unit tests with hand-computed prices cover each new rule. Arms:

- two control replicates;
- placebo;
- template coach;
- two post-test replicates for each of the four coaches.

The junior’s errors. Its practice accuracy fell to 44 of 66, but the pre-registered mutation library explained only **2 of its 22 errors**. Every coach found what the library missed. Without being told, Haiku, Sonnet, Opus and Fable each identified the same systematic misconception: the junior treated the heavy-kg uplift as a separate add-on, so it computed the weekend percentage and the GOLD discount on the ordinary base only. Opus and Fable also caught the secondary errors in two items: a missed rounding to 2 kg, a skipped discount, an omitted remote fee.

We added the coaches’ misconception to the library as a labelled post-hoc entry. It explains **18 of 22** practice errors. Here the models under test out-diagnosed the benchmark’s own authors, which is the construct DEBRIEF claims to measure.

Fresh-case scores. Control accuracy on the fresh cases was 87.8% (replicates 91.1% and 84.4%). The two replicates agreed on only 80% of cases, and the replicate-versus-replicate “NFR” was –75%.

Arm (fresh cases, 90 per run)	Run 1 accuracy	Run 1 NFR	Run 2 accuracy	Run 2 NFR	Pooled NFR [95% CI]
Control	91.1%	—	84.4%	—	—
Placebo note	90.0%	+18	—	—	+18 [−50, 100]
Template coach	87.8%	0	—	—	0 [−100, 100]
Haiku coach	95.6%	+64	92.2%	+36	+50 [8, 76]
Sonnet coach	96.7%	+73	90.0%	+18	+45 [20, 92]
Opus coach	98.9%	+91	44.4%	−355	−132 [−327, −55]
Fable coach	93.3%	+45	76.7%	−91	−23 [−157, 40]

In the first post-test replicate, every coach beat both baselines. Opus’s notes lifted the junior to 98.9% accuracy (NFR +91, 95% CI [76, 100]). The second replicate of the same notes collapsed for Opus and Fable, for two reasons:

- **A harness artefact.** Items k and $k+3$ shared a seed, so each L4 tariff had a near-identical L5 twin. Agents that read all six prompts in one context sometimes applied the twin’s amendments, and one wrote another item’s answers into a file. Screening out the five affected item-runs (post hoc) raises pooled NFR to +14 for Opus and +9 for Fable.
- **Note-induced over-generalisation.** On the remaining items the junior applied a document cap mentioned in a note to parcels. These are exactly the “breaks” DEBRIEF is designed to subtract, but they appeared in one replicate and not the other for the same note.

Gates.

Gate	Requirement	Outcome
G1	Junior control accuracy 30–70%	Failed (87.8%)
G2	Practice errors diagnosable by the library	Failed as pre-registered (2/22); passes with the coach-discovered entry (18/22)
G3	Template coach beats placebo	Failed (0 vs +18; the template had nothing to blame)
G4	Placebo within ± 5 points of 0	Not met (+18, CI spans −50 to 100)
G6	Junior repeatable enough to rank coaches	Failed: the same note produced NFR +91 and −355

5.3 What the DEBRIEF pilot shows

The construct is real. Four models diagnosed a misconception that the pre-registered library missed, wrote notes under 150 words, and in their first replicates removed 45 to 91% of the junior’s errors on cases they never saw, against 0 to 18% for the baselines. Weaker coaches wrote shorter, safer notes and showed smaller swings.

The instrument is not yet reliable enough to rank models. Variance in how the junior reads a note dominates differences between coaches, and our agent harness added cross-item contamination. These problems are fixable, and the fixes are concrete:

- one item per junior call, at fixed temperature through an API;
- at least 4 replicates;
- at least 60 items with distinct tariffs and no shared seeds;
- several interacting misconceptions per item, so that diagnosis and triage, not just the one dominant fix, separate coaches;
- a pinned junior whose control accuracy sits at 40 to 70%.

With pooled NFR swinging by more than 100 points between replicates, we make no claim that any coach is better than another.

6 Limitations

- **One lab’s models.** All pilot solvers, coaches and juniors were Claude models, run as Claude Code agents with honour-system tool policies rather than a frozen harness. One DEBRIEF coach used shell commands to count words; no coach or junior is known to have run code to compute prices.
- **Tiny samples:** 8 to 60 items per pilot.
- **No human participants.** DEBRIEF’s human-coach and human-learner validation arms remain to be run.
- **Interrupted runs.** Usage limits cut several runs short, so the Season Forge snapshots for Sonnet, Fable and Opus and three Unrun Lab solvers are incomplete.
- **Weaker sourcing.** Several primary hosts (arxiv.org, arcprize.org, epoch.ai, openai.com and others) were blocked, so some claims rest on search summaries [S].
- **Imperfect blinding.** One Season Forge passphrase was written to a local tool-results file, so sealed-file access cannot be ruled out for that pilot.

7 Conclusion

Looking for “a benchmark AI cannot have trained on” targets the wrong property. Every fixed target is trained on within months once it matters. What survives is renewal built into the mechanism: a new game every season, new human authors every month, new rule systems every window. Scores should be anchored to things that can be raised, and should be measured on outcomes other than the model’s own answer. Equally important is willingness to fail one’s own gates: three of five pilots in the search, and DEBRIEF’s first version, failed theirs, and each failure showed where the real difficulty is.

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